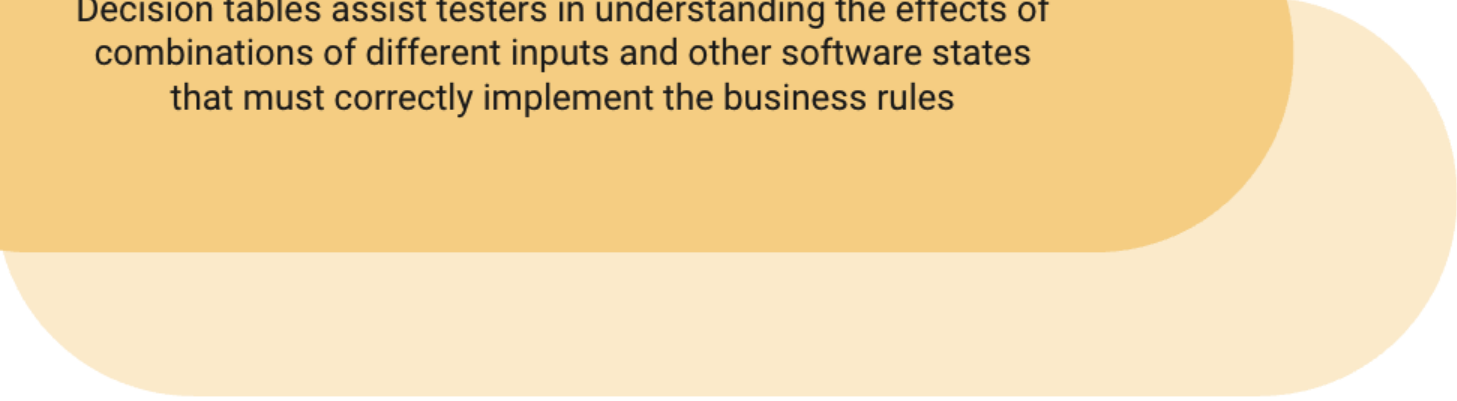
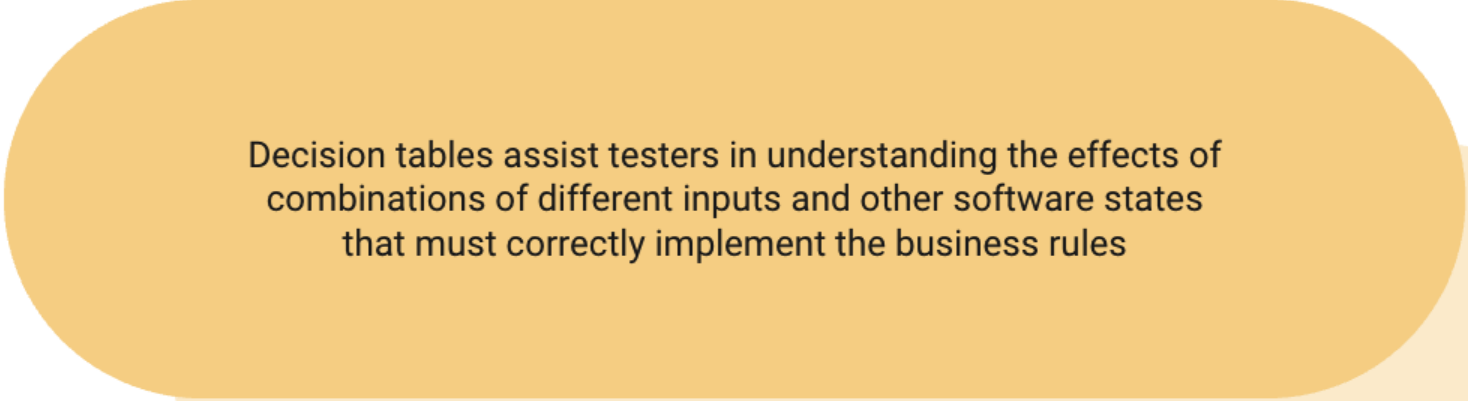


SOFTWARE TESTING

MASTERCLASS

Decision Table and State Transition Testing



Decision tables assist testers in understanding the effects of combinations of different inputs and other software states that must correctly implement the business rules





Advance Testing Concepts

Decision Table and State Transition Testing

Requirements:

- The user should have a positive account balance in order to transfer money to another user
- The user should not be banned.



Advance Testing Concepts

Decision Table and State Transition Testing

Condition	Step 1	Step 2	Step 3	Step 4
User has positive account balance	Y	Y	N	N
User not banned	Y	N	Y	N
Expected Outcome				
Money Transfer Successful	Y	N	N	N



Advance Testing Concepts

Decision Table and State Transition Testing

- **Test Case 1:** User with a positive amount and not banned: Money transfer is allowed.
- **Test Case 2:** User with a positive amount and banned: Money transfer is not allowed.
- **Test Case 3:** User with a negative amount and not banned: Money transfer is not allowed.
- **Test Case 4:** User with a negative amount and banned: Money transfer is not allowed.

Condition	Step 1	Step 2	Step 3	Step 4
User has positive account balance	Y	Y	N	N
User not banned	Y	N	Y	N
Expected Outcome				
Money Transfer Successful	Y	N	N	N





Advance Testing Concepts

Decision Table and State Transition Testing

The number of cases depends on the number of conditions, you can calculate the number of test cases:

Number of Test Cases = 2^n

For 2 condition: $2^2 = 4$ test cases

For 3 condition: $2^3 = 8$ test cases

For 4 condition: $2^4 = 16$ test cases

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Advance Testing Concepts

Decision Table and State Transition Testing

State transition testing aims to ensure that the system transitions between different states correctly and consistently

It is used when features of a system are represented as states that transform into one another



Advance Testing Concepts

Decision Table and State Transition Testing

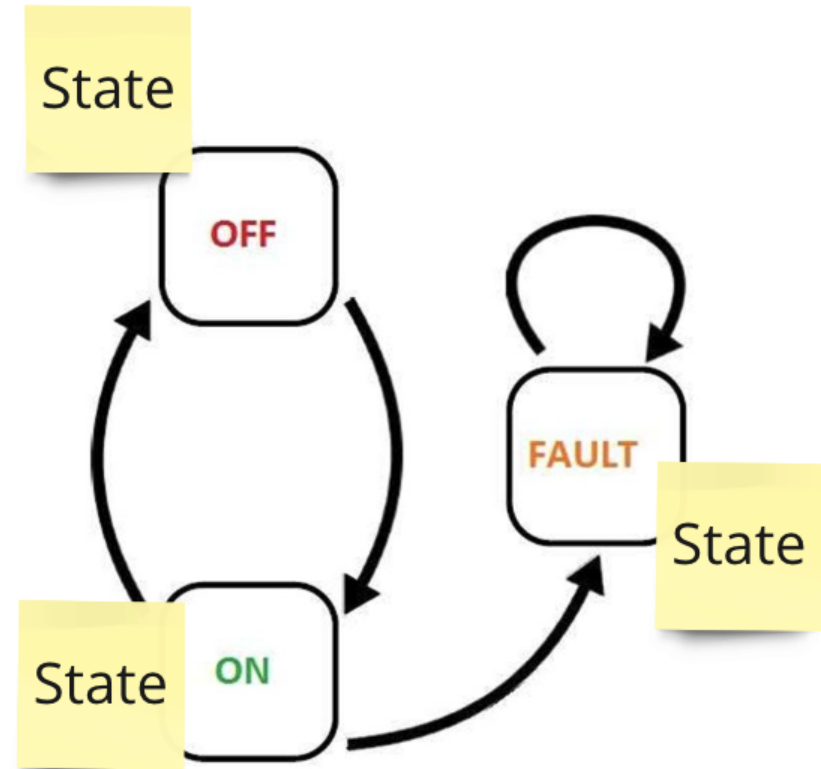


We are testing an electric switch with three states: OFF, ON, and FAULT.

Advance Testing Concepts

Decision Table and State Transition Testing

- **Test Case 1:** Switch on the device when it's off.
- **Test Case 2:** Switch off the device when it's on
- **Test Case 3:** Switch on the device when it's already on, resulting in the FAULT state
- **Test Case 4:** Attempt to switch on the device when it's in the FAULT state, which should remain in the FAULT state



Advance Testing Concepts

Decision Table and State Transition Testing

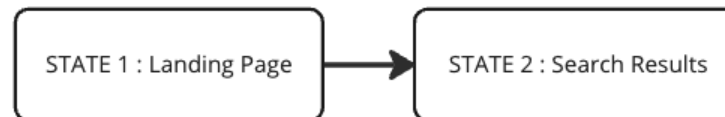


Sequence of events needs to be tested

Application is dependent upon the event value

There are conditions based on the past states

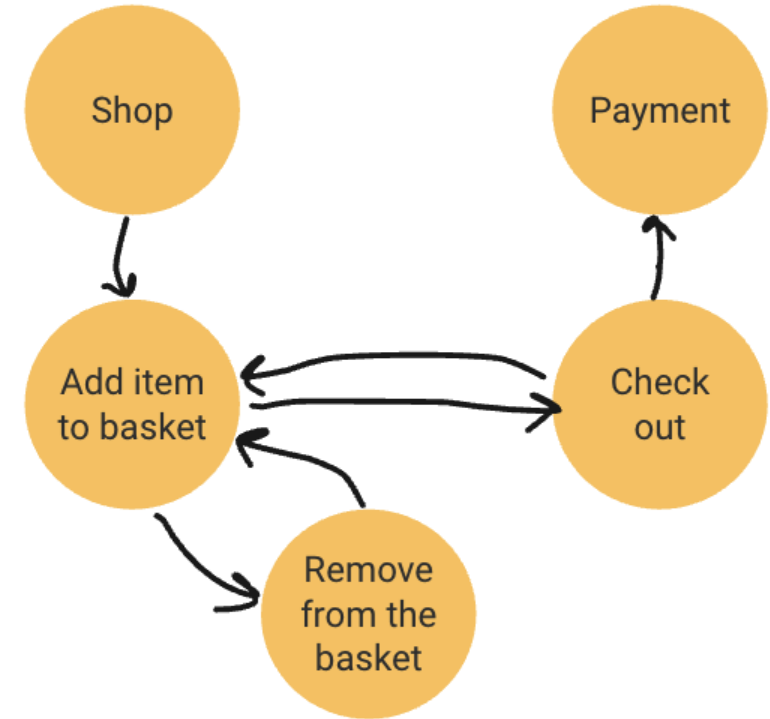
As a user, you perform a search and observe the search results.



All states coverage:

- The coverage items are the **states**.
- To achieve 100% coverage of all states, test cases must ensure that all the states are visited.

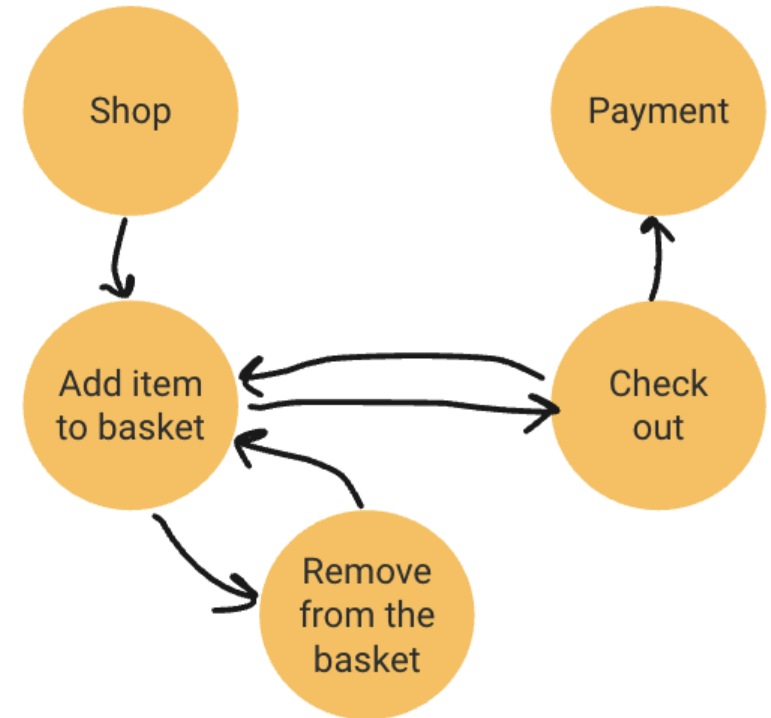
Amazon.com Checkout Flow



Valid transitions coverage / 0-switch coverage:

- "Switch" refers to the act of transitioning from one state to another state in a system
- The coverage items are the **single valid transitions**
- To achieve 100% coverage of valid transitions, test cases must exercise all the valid transitions.

Amazon.com Checkout Flow



All transitions coverage:

- To achieve 100% coverage of all transitions, test cases must exercise all the valid transitions and also attempt to execute invalid transitions
- This criterion ensures comprehensive testing of the state transitions in the system

Amazon.com Checkout Flow

